

Q1: $150 - \{60 - [25 + (18 - 7)]\} = ?$

- (A) 116 (B) 126 (C) 128 (D) 136 (E) 120

$$\Rightarrow 150 - \{60 - [25 + 11]\}$$

$$\Rightarrow 150 - \{60 - 36\}$$

$$\Rightarrow 150 - 24 \Rightarrow 126$$

Q2: $37\% \text{ of } 8100 - 46^2 + 5 = ?$

- (A) 896 (B) 876 (C) 886 (D) 906 (E) None of these

$$\Rightarrow (37\% \text{ of } 8100) - 46^2 + 5$$

$$\Rightarrow 37 \times 81 - 46^2 + 5$$

$$\Rightarrow (37 \times 3 \times 27) - 2116 + 5 \Rightarrow 2997 - 2116 + 5 = \underline{886}$$

Q3: $250 - \{(1.44 \div 0.12) + (0.21 \div 0.03)\} = ?$

- (A) 233 (B) 229 (C) 231 (D) 241 (E) None of these

$$\Rightarrow 250 - \left\{ \left(\frac{144}{100} \div \frac{12}{100} \right) + \left(\frac{21}{100} \div \frac{3}{100} \right) \right\}$$

$$\Rightarrow 250 - \left\{ \left(\frac{144}{100} \times \frac{100}{12} \right) + \left(\frac{21}{100} \times \frac{100}{3} \right) \right\}$$

$$\Rightarrow 250 - \{12 + 7\} \Rightarrow 250 - 19 \Rightarrow 231$$

Q4: $(24 \times 35 \times 6) - (42 \times ?) = 3528$

- (A) 40 (B) 48 (C) 32 (D) 36 (E) None of these

$$(24 \times 35 \times 6) - (42 \times x) = 3528$$

$$120 - x = 84$$

$$x = 120 - 84$$

$$= 36$$

{ Divided both sides
by 42 }

Q5: $(8^9 \div 8^4) \div 8^2 = ?$

- (A) 64 (B) 512 (C) 128 (D) 256 (E) None of these

$$\Rightarrow (8^9 \div 8^4) \div 8^2$$

$$\Rightarrow 8^{(9-4)} \div 8^2$$

$$\Rightarrow 8^5 \div 8^2 \Rightarrow 8^{(5-2)} \Rightarrow 8^3$$

$$= 512$$

Q6: $\sqrt[3]{1331} + \sqrt{1024} + \sqrt[4]{625} = ? + \sqrt[3]{216}$

(A) 32

(B) 44

(C) 42

(D) 40

(E) None of these

$$(1331)^{\frac{1}{3}} + (1024)^{\frac{1}{2}} + (625)^{\frac{1}{4}} = x + (216)^{\frac{1}{3}}$$

$$(11^3)^{\frac{1}{3}} + (32^2)^{\frac{1}{2}} + (5^4)^{\frac{1}{4}} = x + (6^3)^{\frac{1}{3}}$$

$$11 + 32 + 5 = x + 6 \Rightarrow x = 42$$

Q7: $\frac{1456 + 972 - 1128 + 1700}{25} = ?$

(A) 115

(B) 125

(C) 130

(D) 120

(E) None of these

$$\Rightarrow \frac{1456 + 972 - 1128 + 1700}{25} \Rightarrow \frac{2428 - 1128 + 1700}{25}$$

$$\Rightarrow \frac{1300 + 1700}{25} = \frac{3000}{25} = \frac{3000}{100} \times 4 \Rightarrow 120$$

Q8: $(? + 24)\% \text{ of } 1200 = 996$

(A) 55

(B) 59

(C) 62

(D) 60

(E) None of these

$$(x + 24)\% \times 1200 = 996$$

$$x + 24 = \frac{996}{12}$$

$$x + 24 = 83 \Rightarrow x = 59$$

Q9: $[(7.25 - 3.85) \div 0.17] + (0.08 \times 375) = ?$

(A) 52

(B) 48

(C) 46

(D) 50

(E) None of these

$$\Rightarrow [(7.25 - 3.85) \div 0.17] + (0.08 \times 375)$$

$$\Rightarrow [(4 - 0.60) \div 0.17] + (0.08 \times 375)$$

$$\Rightarrow [3.40 \div 0.17] + \left[\frac{8}{100} \times 375\right] \Rightarrow 20 + 30 = 50$$

Q10: $125 + 1024 - 343 + ?^2 = 1767$

(A) 31

(B) 29

(C) 33

(D) 35

(E) None of these

$$125 + 1024 - 343 + x^2 = 1767$$

$$1149 + x^2 = 1767 + 343$$

$$x^2 = 2110 - 1149$$

$$x^2 = 961$$

$$x = 31$$

Q11: $500 - \{60\% \text{ of } [180 \div (5 - 2) - (20\% \text{ of } 150)]\} = ?$

(A) 480

(B) 484

(C) 482

(D) 488

(E) None of these

$$\Rightarrow 500 - \left\{ 60\% \text{ of } \left[\left(\frac{180}{3} \right) - \left(\frac{20}{100} \times 150 \right) \right] \right\}$$

$$\Rightarrow 500 - \left\{ \frac{60}{100} \times (60 - 30) \right\}$$

$$\Rightarrow 500 - 18 \Rightarrow 482$$

Q12: $6\frac{1}{5} - 2\frac{3}{10} + 4\frac{7}{8} = 7\frac{3}{20} + ? + 1\frac{1}{4}$

(A) 5/8

(B) 3/4

(C) 7/8

(D) 1/8

(E) None of these

$$6 + \frac{1}{5} - 2 - \frac{3}{10} + 4 + \frac{7}{8} = 7 + \frac{3}{20} + x + 1 + \frac{1}{4}$$

$$(6 - 2 + 4 - 7 - 1) + \frac{1}{5} - \frac{3}{10} + \frac{7}{8} - \frac{3}{20} - \frac{1}{4} = x$$

$$0 + \left(\frac{8 - 12 + 35 - 6 - 10}{40} \right) = x \Rightarrow x = \frac{15}{40} \text{ or } \frac{3}{8}$$

Q13: $(18.75\% \text{ of } 960) \times (6.25\% \text{ of } ?) = 1800$

(A) 140

(B) 150

(C) 180

(D) 160

(E) None of these

$$\left(\frac{3}{16} \times 960 \right) \times \left(\frac{1}{16} \times x \right) = 1800$$

$$180 \times \frac{x}{16} = 1800 \Rightarrow x = \frac{1800 \times 16}{180}$$

$$x = 160$$

Q14: $? + \frac{11}{12} - \frac{7}{18} + \frac{5}{9} = 15\frac{3}{4} + 13\frac{5}{6} - 1\frac{1}{2}$

(A) 27

(B) 26

(C) 25

(D) 28

(E) None of these

$$x + \frac{11}{12} - \frac{7}{18} + \frac{5}{9} = 15 + \frac{3}{4} + 13 + \frac{5}{6} - 1 - \frac{1}{2}$$

$$x = (15 + 13 - 1) + \frac{3}{4} + \frac{5}{6} - \frac{1}{2} - \frac{11}{12} + \frac{7}{18} - \frac{5}{9}$$

$$x = 27 + \frac{27 + 30 - 18 - 33 + 14 - 20}{36} \Rightarrow 27 + 0 = 27$$

Q15: $31^2 + (45^2 \div 15) + 22\% \text{ of } 950 = ?$

(A) 1315

(B) 1305

(C) 1325

(D) 1295

(E) None of these

$$961 + \left(\frac{45 \times 45}{15} \right) + \frac{11}{100} \times 1900$$

$$961 + 135 + 209$$

$$= 1305$$

$$\left\{ \begin{array}{l} 22\% \text{ of } 950 \\ \left(\frac{22\%}{2} \right) \text{ of } (950) \times 2 \\ = 11\% \text{ of } 1900 \end{array} \right\}$$

$$\left\{ \begin{array}{l} 22\% \text{ of } 950 \\ \left(\frac{22\%}{2} \right) \text{ of } (950) \times 2 \\ = 11\% \text{ of } 1900 \end{array} \right\}$$

$$= 11\% \text{ of } 1900$$

Q16: $(0.4)^2 = 0.004096$

(A) 4

(B) 8

(C) 5

(D) 7

(E) None of these

$$\left(\frac{4}{10}\right)^x = \left(\frac{4096}{1000000}\right) \Rightarrow \frac{4^x}{10^x} = \left(\frac{4}{10}\right)^6 = (0.4)^6$$

Q17: $(1 \div 3.125\%) + (1 \div 12.5\%) = ?$

(A) 32

(B) 40

(C) 36

(D) 48

(E) None of these

$$\Rightarrow \frac{1}{1/32} + \frac{1}{1/8}$$

$$\Rightarrow 32 + 8$$

$$\Rightarrow 40$$

$$\left. \begin{aligned} 6 \cdot 25\% &= \frac{1}{16} \\ 31 \cdot 25\% &= 6 \cdot 25\% \times 5 = \frac{5}{16} \\ 3 \cdot 25\% &= \frac{5}{16} \times \frac{1}{5} = \frac{1}{32} \end{aligned} \right\}$$

Q18: $60 \div \left[\frac{1}{12} + \frac{11}{12} \times \frac{3}{8} \div (3 \div 2^5) \right] = ?$

(A) 12

(B) 16

(C) 8

(D) 20

(E) None of these

$$60 \div \left[\frac{1}{12} + \left(\frac{11}{12} \times \frac{3}{8} \div \frac{3}{32} \right) \right]$$

$$60 \div \left[\frac{1}{12} + \left(\frac{11}{12} \times \frac{3}{8} \times \frac{32}{3} \right) \right] \text{ \{ Reciprocal \}}$$

$$60 \div \left(\frac{1}{12} + \frac{11}{3} \right) \Rightarrow 60 \div \frac{45}{12} \Rightarrow 60 \times \frac{12}{45} \Rightarrow 16$$

Q19: $? \% \text{ of } 360 = 30^2 - 36 \times 23$

(A) 25

(B) 20

(C) 15

(D) 30

(E) None of these

$$x \% \text{ of } 360 = (30 \times 30) - (36 \times 23)$$

$$x \% \text{ of } 10 = 25 - 23$$

$$\frac{x}{100} \times 10 = 2$$

$$x = 20$$

$$\left\{ \begin{aligned} &\text{Divided both} \\ &\text{sides by 36} \end{aligned} \right\}$$

Q20: $11^2 + 2^{10} - 5^4 = ? - 14^2$

(A) 724

(B) 712

(C) 716

(D) 720

(E) None of these

$$11^2 + 2^{10} - 5^4 = x - 14^2$$

$$121 + 1024 - 625 = x - 196$$

$$1145 - 625 + 196 = x$$

$$520 + 196 = x$$

$$716 = x$$

$$0.2^1 = 0.2$$

$$0.2^2 = 0.04$$

$$0.2^3 = 0.008$$

$$0.2^4 = 0.0016$$

$$0.3^1 = 0.3$$

$$0.3^2 = 0.09$$

$$0.3^3 = 0.027$$

$$0.3^4 = 0.0081$$

Q26: $(0.2)^{12} \times (0.04)^5 \times (0.008)^2 = (0.0016)^{?+3}$

(A) 3

(B) 2

(C) 5

(D) 4

(E) None of these

$$(0.2)^{12} \times (0.2^2)^5 \times (0.2^3)^2 = (0.2^4)^{x+3}$$

[Let $0.2 = a$]

$$a^{12} \times a^{10} \times a^6 = a^{4x+12}$$

$$a^{(12+10+6)} = a^{(4x+12)}$$

$$a^{28} = a^{4x+12}$$

$$4x+12 = 28$$

$$4x = 16$$

$$x = 4$$

Q27: $\left(\left(\frac{4}{9} \times 2\right) + \frac{7}{9}\right)$ of $66.67\% - 100\% = ?$

(A) 12.5%

(B) 7.69%

(C) 14.29%

(D) 11.11%

(E) None of these

$$\left[\left(\frac{8}{9} + \frac{7}{9}\right) \times \frac{2}{3}\right] - 1$$

$$\left[\frac{15}{9} \times \frac{2}{3}\right] - 1 \Rightarrow \frac{10}{9} - 1 \Rightarrow \frac{1}{9} \text{ or } 11.11\%$$

Q28: $7 \times 7 \times 7 \times 7 \times 7 \times 7 + 7 \times 7 \times 7 \times 7 \times 7 = 196 \times 3.5 \times ?$

(A) 196

(B) 182

(C) 224

(D) 168

(E) None of these

$$(7 \times 7 \times 7 \times 7 \times 7 \times 7) + (7 \times 7 \times 7 \times 7 \times 7) = 196 \times 3.5 \times x$$

$$7^6 + 7^5 = 196 \times 3.5 \times x$$

$$7^5(7+1) = 98 \times 7 \times x$$

$$7^5 \times 8 = 2 \times 7^3 \times x \Rightarrow$$

$$7^2 \times 4 = x$$

$$x = 49 \times 4$$

$$x = 196$$

Q29: $14\% \text{ of } 9800 - 21\% \text{ of } 4900 = ? \times 7^2$

(A) 5

(B) 9

(C) 7

(D) 11

(E) None of these

$$(14\% \text{ of } 9800) - (21\% \text{ of } 4900) = x \times 7^2$$

$$(28\% \text{ of } 4900) - (21\% \text{ of } 4900) = x \times 7^2$$

$$4900(28\% - 21\%) = x \times 7^2$$

$$4900 \times \frac{7}{100} \times \frac{1}{49} = x$$

$$x = 7$$

Q21: 82.5% of 320 - 17.5% of 240 = ?

(A) 210

(B) 232

(C) 242

(D) 222

(E) None of these

$$\Rightarrow \left(\frac{33}{40} \times 320 \right) - \left(\frac{7}{40} \times 240 \right)$$

$$\Rightarrow 264 - 42$$

$$\Rightarrow 222$$

Q22: ? = $(14^2 - 11^2) \times (25^2 - 15^2) \div (17.5^2 - 2.5^2)$

(A) 104

(B) 96

(C) 98

(D) 100

(E) None of these

$$x = \frac{[(14+11)(14-11)] \times [(25+15)(25-15)]}{(17.5+2.5)(17.5-2.5)}$$

$$x = \frac{25 \times 3 \times 40 \times 10}{20 \times 15} \Rightarrow x = 100$$

Q23: 6.25% of $\left(\frac{3}{5} + \left(\frac{1}{5} \times \frac{1}{5}\right)\right) = ?$

(A) 5%

(B) 6%

(C) 3%

(D) 7%

(E) None of these

$$\frac{1}{16} \text{ of } \left(\frac{3}{5} + \frac{1}{25} \right) \Rightarrow \frac{1}{16} \times \left(\frac{15+1}{25} \right)$$

$$\Rightarrow \frac{1}{16} \times \frac{16}{25} = \frac{1}{25} \text{ or } 4\%$$

Q24: $\left[\left(\frac{4}{7} \right) \div \left\{ 1 - \left(\frac{3}{7} \right) \right\} \right] \times \left[\left(\frac{9}{10} \right) \div \left\{ 1 - \left(\frac{2}{5} \right) \right\} \right] = ?$

(A) 2/3

(B) 5/2

(C) 3/5

(D) 1/2

(E) None of these

$$\Rightarrow \left[\frac{4}{7} \div \frac{4}{7} \right] \times \left[\frac{9}{10} \div \frac{3}{5} \right]$$

$$\Rightarrow 1 \times \frac{9}{10} \times \frac{5}{3}$$

$$\Rightarrow 1 \times \frac{3}{2} \Rightarrow \frac{3}{2}$$

Q25: $(7.5\% \text{ of } 2000) \times (8\frac{1}{3}\% \text{ of } 600) - 12.5\% \text{ of } 24000 = ?$

(A) 4500

(B) 4800

(C) 5000

(D) 4200

(E) None of these

$$\left(\frac{3}{40} \times 2000 \right) \times \left(\frac{1}{12} \times 600 \right) - \left(\frac{1}{8} \times 24000 \right) = x$$

$$(150 \times 50) - 3000$$

$$7500 - 3000 = 4500$$

$$\left. \begin{aligned} 17.5\% &= \frac{7}{40} \\ \text{and} \\ 82.5\% &= 100\% - 17.5\% \\ &= 1 - \frac{7}{40} = \frac{33}{40} \end{aligned} \right\}$$

$$\left\{ A^2 - B^2 = (A+B)(A-B) \right\}$$

$$\left\{ 1 - \frac{3}{7} = \frac{7-3}{7} = \frac{4}{7} \right\}$$

$$\left[\begin{aligned} 8\frac{1}{3}\% &= 8.33\% \\ \text{and} \\ 8.33\% &= \frac{1}{12} \end{aligned} \right]$$